

PRODUCTIVITY & CHROME EXTENSION PLATFORM

Project Overview

Student productivity platform blocking distracting websites, featuring YouTube Study Mode, focus session timers, streaks & gamified leaderboards.

- Problem Statement:**

Students lose study hours to social media notifications and algorithmic YouTube recommendations during critical study sessions.
- Solution Implemented:**

Built a Chrome Manifest V3 extension & web platform enforcing scheduled website blocks, YouTube feed removal, and focus telemetry.
- My Technical Role:**

Lead Full-Stack Developer & Chrome Extension Architect

Main Features & Technical Stack

- Core Features:**

 - Scheduled site blocklist & custom block rules
 - YouTube Study Mode stripping recommended video feeds
 - Focus Pomodoro timer with automated history logging
 - Gamified student focus streaks and leaderboards
- Technologies Used:**

React, Vite, FastAPI, Python, SQLite, SQLAlchemy, Chrome Manifest V3, Vercel
- Architecture Flow:**

React / Extension Popup → FastAPI (Python) API Gateway → JWT Auth → SQLite DB → Analytics Engine

Technical Challenge & Results

- Key Engineering Challenge:** Manifest V3 background worker lifecycle termination fixed using persistent Alarm APIs & offscreen documents.
- Validated Results:** Proven zero false-positive site blocks and hundreds of recorded deep-work focus hours.

Live Demo: <https://github.com/prashant-singh-78/SWD.git> | **GitHub:** <https://github.com/prashant-singh-78/SWD.git> | **Portfolio:** prashant08.vercel.app